

Electric test vehicle: data sheet

Measure what is marked required; the rest have defaults that work until you measure them. Enter the values in a copy of configs/datasheets/template.yaml named after your vehicle and run python scripts/build_vehicle.py configs/datasheets/<your_name>.yaml. Positions are measured from the top (nose) of the vehicle downward. Write each measured value in the Value column, then copy it to the entry of the same name in the sheet. docs/vehicle_data_request.md and the template say how to measure each one.

Filled in by: _____ Date: _____

Vehicle (vehicle:)

	What	Entry	Unit	Required or default	Value
☐	Short name for the config files: letters, digits, _	name	-	the sheet's file name	
☐	Mass ready to fly, battery and fan unit in	total_mass_g	g	required	
☐	Balance point (centre of gravity) from the top, ready to fly	balance_point_mm	mm	required	
☐	Body length from the top to the bottom end, legs not counted	length_mm	mm	required	
☐	Body outside diameter	diameter_mm	mm	required	
☐	Pitch inertia of the whole vehicle about its balance point	pitch_inertia_kgm2	kg m2	solid cylinder	
☐	Roll inertia of the whole vehicle about its long axis	roll_inertia_kgm2	kg m2	solid cylinder	
☐	Drag coefficient on the body diameter	drag_coefficient	-	0.8	
☐	Centre of pressure from the top, wind from the side	centre_of_pressure_mm	mm	middle of the body	
☐	Side force slope for wind across the body	side_force_slope_per_rad	1/rad	8	

Fan and battery (fan:)

	What	Entry	Unit	Required or default	Value
☐	Fan, motor and ESC model	model	-	optional	
☐	Fan, motor and ESC as mounted, battery not included	unit_mass_g	g	required	
☐	Centre of the fan unit from the top	position_mm	mm	required	
☐	Fan (duct) diameter	diameter_mm	mm	required	
☐	Fan unit length	length_mm	mm	same as the diameter	
☐	Static thrust at full throttle, freshly charged battery	max_thrust_n	N	required	
☐	Static thrust at 50 % throttle	half_throttle_thrust_n	N	not checked	
☐	Time for the thrust to reach 63 % of a throttle step	spin_up_ms	ms	required	
☐	Hover time on one charge down to the low voltage cut	battery_time_s	s	required	
☐	Roll torque of the spinning fan per newton of thrust	reaction_torque_nm_per_n	N m/N	0.003	

Thrust vectoring and servos (thrust_vectoring:)

	What	Entry	Unit	Required or default	Value
☐	Gimbal pivot or vane hinge line from the top	pivot_mm	mm	required	
☐	Largest thrust deflection each way, each axis	max_angle_deg	deg	required	
☐	Thrust angle per degree of servo horn travel	linkage_ratio	-	required	
☐	Servo model	servo_model	-	optional	
☐	Servo speed: time for 60 deg of horn travel at your voltage	servo_time_per_60deg_s	s	required	
☐	Time from a command to the horn starting to move	servo_delay_ms	ms	20	
☐	Smallest thrust angle change the servo follows	servo_deadband_deg	deg	0.1	
☐	Pulse that holds the thrust straight	servo_center_us	us	1500	
☐	Pulse change per degree of horn travel	servo_us_per_deg	us/deg	10	
☐	Shortest pulse the flight code may send	servo_min_us	us	1000	
☐	Longest pulse the flight code may send	servo_max_us	us	2000	
☐	Smallest pulse step the board puts out	pulse_resolution_us	us	1	
☐	+1 if a longer pitch servo pulse swings the air jet toward +x, else -1	pitch_servo_sign	-	1	
☐	+1 if a longer yaw servo pulse swings the air jet toward +y, else -1	yaw_servo_sign	-	1	

Roll control (roll_control:)

	What	Entry	Unit	Required or default	Value
☐	Roll fans, reaction wheel or vanes	kind	-	optional	
☐	Largest roll torque	max_torque_nm	N m	required	
☐	Time to reach 63 % of a torque step	response_ms	ms	50	

Landing legs (legs:)

	What	Entry	Unit	Required or default	Value
☐	Number of legs	count	-	required	
☐	Diameter of the circle through the feet	foot_circle_mm	mm	required	
☐	How far the feet reach below the bottom end of the body	below_body_mm	mm	required	
☐	Highest drop onto the legs they survive, at flight mass	max_drop_height_mm	mm	required	
☐	Sideways speed at touchdown the legs survive	max_sideways_speed_mps	m/s	1	
☐	Tilt at touchdown the legs survive	max_tilt_deg	deg	10	

Sensors and flight computer (sensors :)

	What	Entry	Unit	Required or default	Value
☐	Flight computer board	flight_computer	-	optional	
☐	How often the flight code runs the control loop	control_rate_hz	Hz	required	
☐	Control steps from reading the sensors to the servo command	command_delay_steps	steps	1	
☐	IMU model	imu_model	-	optional	
☐	IMU sample rate the flight code reads	imu_rate_hz	Hz	required	
☐	IMU filter delay	imu_lag_ms	ms	2	
☐	Accelerometer noise	accel_noise_mps2	m/s2	0.05	
☐	Accelerometer bias change between power-ups	accel_bias_mps2	m/s2	0.02	
☐	Accelerometer full scale (16 g is 157)	accel_range_mps2	m/s2	156	
☐	Gyroscope noise	gyro_noise_deg_per_s	deg/s	0.1	
☐	Gyroscope bias change between power-ups	gyro_bias_deg_per_s	deg/s	0.5	
☐	Gyroscope full scale	gyro_range_deg_per_s	deg/s	2000	
☐	Barometer model	barometer_model	-	optional	
☐	Barometer sample rate the flight code reads	barometer_rate_hz	Hz	required	
☐	Barometer filter delay	barometer_lag_ms	ms	20	
☐	Barometer height noise	barometer_noise_m	m	0.3	
☐	Barometer height drift over a flight	barometer_bias_m	m	1	
☐	GPS model	gps_model	-	optional	
☐	GPS update rate, 0 if there is no GPS	gps_rate_hz	Hz	required	
☐	GPS delay	gps_lag_ms	ms	100	
☐	GPS position scatter from sample to sample	gps_noise_m	m	0.8	
☐	Slow GPS position wander over a minute	gps_drift_m	m	1	
☐	GPS speed noise	gps_speed_noise_mps	m/s	0.1	

Launch site (launch_site :)

	What	Entry	Unit	Required or default	Value
☐	Field height above sea level	elevation_m	m	300	
☐	Steady wind expected on test day	wind_mps	m/s	4	
☐	Gust strength (standard deviation)	gust_mps	m/s	1.5	

Mission (mission:)

	What	Entry	Unit	Required or default	Value
☐	Hover height of the feet above the pad	target_height_m	m	required	
☐	Hover time at the target height	hover_time_s	s	required	
☐	Touchdown must be within this distance of the pad	landing_radius_m	m	required	
☐	Counts as at the target height within this band	height_tolerance_m	m	2	
☐	Planned climb speed	climb_speed_mps	m/s	4	
☐	How quickly the plan speeds up and slows down	climb_acceleration_mps2	m/s2	1.5	
☐	Planned descent speed	descent_speed_mps	m/s	3	
☐	Height where the plan slows to the touchdown speed	final_height_m	m	4	
☐	Touchdown speed the plan aims for	final_speed_mps	m/s	0.7	
☐	Extra hover time planned so the required time is met with margin	extra_hover_s	s	2	
☐	Flight time limit	max_flight_time_s	s	plan + 40 s	
☐	Tilt at which the flight computer cuts the fan	abort_tilt_deg	deg	40	
☐	Distance from the pad at which the flight computer cuts the fan	geofence_radius_m	m	40	